



Fig. 2: AM4096 magnetic encoder chip from Renishaw. All of the sensor and processing electronics have been placed within the compact silicon design. The rotation of a simple north/south magnet is picked up by the AM4096's sensor and provides absolute positional information output to an accuracy of better than 0.1 degree

Note that a variant of VR sensor is available as inductive magnetic sensor. It differs from the standard VR sensor in that the magnet is not in the unit, but in the object being sensed. Inductive magnetic sensors are used when the rotating device sensed cannot be made of ferrous materials.

Magnetostrictive speed sensor. Rotational speed measurement using a magnetostrictive (MR) sensor is achieved by counting ferromagnetic marks, such as teeth of a passive gear wheel or the number of magnetic elements of a magnetised ring. Magnetostrictive sensors make use of the magnetostrictive effect—the property of a current-carrying magnetic material to change its resistivity in the presence of an external magnetic field. Although inductive sensors can be used for this task, magnetostrictive sensors have an advantage that the output signal level does not vary with rotation speed, as in the case of inductive sensors.

MR sensors are static and the output signal is generated by the bending of magnetic field lines according to the position of the target wheel. As bending of the magnetic field lines also occurs when the target is not moving, these sensors can measure very slow rotations, even down to zero hertz (0 Hz). However, note that MR sensors cannot directly measure rotational speed, but are sensitive to the motion of a toothed wheel made from ferrous material (passive target, a sensor fitted with a permanent magnet) or a rotating wheel having alternating magnetic poles (active target).

RPM measurement techniques

Frequency measurement and period measurement are two widely accepted methods for determining RPM. Frequency measurement is better for fast-moving devices such as motors and turbines, while period measurement is better for devices that move more slowly, such as shafts. When using frequency measurement as a method of monitoring RPM, the key factor is the number of pulses being sensed per revolution (PPR). This method works well with high-PPR sensors and works poorly for low-PPR sensors.

When using frequency measurement method, you can calculate the RPM using this equation:

$$\text{RPM} = (\text{Pulse Frequency in pulses/sec}) \times (60 \text{ sec/min}) / (\text{Sensor pulses/revolution}) = \text{Revolutions/Minute}$$

This equation shows the relationship between frequency and period as: $\text{Frequency} = 1/\text{Period}$

When using period measurement to monitor RPM, the RPM can be calculated by using the equation:

$$\text{RPM} = 60 / (\text{Pulse period} \times \text{PPR})$$

In addition, note that frequency method is good for zero-speed detection. If you use frequency method to monitor the RPM, you can know the shaft has stopped when the frequency being sensed drops to zero. On the other hand, if you use period method, the main issue is determining how much time to allow between pulses before deciding that the shaft has stopped. Basically, this method requires a little more complex logic!

From printing to life care!

Rotational speed sensors are widely used to monitor the speeds of machine components, from printing and paper machines to aircrafts, ships and wind turbines. A tachometer can even find uses in medicine to diagnose circulatory problems such as clogged arteries. By placing a small, turbine-like device called haematachometer in an artery or vein, doctors can use a tachometer to interpret the rate of blood flow from the speed at which the turbine spins.

Nowadays, innovative rotational speed sensor technology offers a wide range of standard products for the most varied applications, with suitable measuring systems and electronics, as well as appropriate housings, designs and connections. The design trend is towards fabrication of tiny chips, i.e., chips integrated with sensor bridges, signal amplifiers, gain controls, differential analogue line drivers and digital interface circuits, for contactless RPM measurement, position control/sensing and rotating angle measurement, etc. ●

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